OUTPUT

```
MODIFIED AVL IMBALANCE DETECTION
Enter number of nodes: 3
Enter 3 values:
30 20 10

First unbalanced ancestor: 30
Balance factor: 2
Imbalance type: LL
Rotation: Right Rotation
Affected nodes: 30 and 20

Inorder Traversal: 10 20 30

Balanced AVL Tree:
     30

20

     10

AVL Balance Verification: PASS - Tree is balanced.

--------------------------------
Process exited after 19.37 seconds with return value 0
Press any key to continue . . .
```

```
MODIFIED AVL IMBALANCE DETECTION
Enter number of nodes: 3
Enter 3 values:
10 20 30

First unbalanced ancestor: 10
Balance factor: -2
Imbalance type: RR
Rotation: Left Rotation
Affected nodes: 10 and 20

Inorder Traversal: 10 20 30

Balanced AVL Tree:
     30

20

     10

AVL Balance Verification: PASS - Tree is balanced.

--------------------------------
Process exited after 21.97 seconds with return value 0
Press any key to continue . . .
```

```
MODIFIED AVL IMBALANCE DETECTION
Enter number of nodes: 3
Enter 3 values:
30 10 20

First unbalanced ancestor: 30
Balance factor: 2
Imbalance type: LR
Rotation: Left Rotation
Affected nodes: 10 and 20
Rotation: Right Rotation
Affected nodes: 30 and 20

Inorder Traversal: 10 20 30

Balanced AVL Tree:
     30

20

     10

AVL Balance Verification: PASS - Tree is balanced.

--------------------------------
Process exited after 19.08 seconds with return value 0
Press any key to continue . . .
```

```
MODIFIED AVL IMBALANCE DETECTION
Enter number of nodes: 3
Enter 3 values:
10 30 20

First unbalanced ancestor: 10
Balance factor: -2
Imbalance type: RL
Rotation: Right Rotation
Affected nodes: 30 and 20
Rotation: Left Rotation
Affected nodes: 10 and 20

Inorder Traversal: 10 20 30

Balanced AVL Tree:
     30

20

     10

AVL Balance Verification: PASS - Tree is balanced.

--------------------------------
Process exited after 12.6 seconds with return value 0
Press any key to continue . . .
```